

NOTES FOR A.S.V. PRESENTATION

Constrained Optimization and Lagrange Multipliers

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THEOREM (Implicit Function Theorem, local homeomorphism version)

Let $U \subset \mathbb{R}^n$ be an open set, and let $g: U \rightarrow \mathbb{R}^m$ be a C^2 function where $1 \leq m < n$.

Write $x \in \mathbb{R}^n$ as $x = (y, z)$, where $y \in \mathbb{R}^{n-m}$ and $z \in \mathbb{R}^m$.

Suppose $x^* = (y^*, z^*) \in U$ satisfies $g(y^*, z^*) = 0$, and the partial $m \times m$ Jacobian matrix $D_z g(y^*, z^*)$ is invertible.

Define the constraint surface

$$S := \{x \in U \mid g(x) = 0\}$$

Then there exists an open neighbourhood $W \subset \mathbb{R}^{n-m}$ of y^* , and an open neighbourhood $N \subset \mathbb{R}^m$ of z^*

and a (unique) C^2 function $\phi: W \rightarrow \mathbb{R}^m$ such that the parametrization mapping $P: W \rightarrow S \cap N$ defined by $P(y) := (y, \phi(y))$ is a homeomorphism.

SETUP

Fix $f \in C^2(U, \mathbb{R})$ where U is open in $\mathbb{R}^n$. Let $g: U \rightarrow \mathbb{R}^m$ be of class C^2 , with $m < n$. Define $S := \{x \in U \mid g(x) = 0\}$.

We seek the local extrema of f restricted to S .

To get suitable necessary as well as sufficient conditions, all require a ~~big~~ local parametrization of the constrained manifold S . So for that we assume henceforth that

$x^* \in S$ is a regular point of g

This means that the $m \times n$ Jacobian matrix $Dg(x^*)$ has full rank m . By permuting the coordinates of g

$$x = (y, z) \in \mathbb{R}^{n-m} \times \mathbb{R}^m,$$

we may assume, WLOG, that the $m \times m$ partial Jacobian

$D_z g(x^*)$ is invertible.

We can now apply the inverse function theorem on g exactly as stated to get

$$y^* \in W \subset \mathbb{R}^{n-m}, \quad x^* \in N \subset \mathbb{R}^n$$

$\phi \in C^2(W, \mathbb{R}^m)$ and a homeomorphism

$P: W \rightarrow S \cap N$ defined by $P(y) = (y, \phi(y))$.

Observe that the given conditions imply that

$$\phi(y^*) = z^*, \text{ or } P(y^*) = x^*.$$

We now show a correspondence b/w the

constrained local minimum of f on S and

the unconstrained local minimum of h on W , where

we define $h: W \rightarrow \mathbb{R}$ by $h(y) := f(P(y))$
 $= f(y, \phi(y)).$

Lemma 1: x^* is a constrained local minimum of f on S if and only if y^* is an unconstrained local minimum of $h = f \circ P$ on W .

Proof: ($\Rightarrow$) Assume x^* is a constrained local minimum of f on S . Then there exists an open neighbourhood $V \subset \mathbb{R}^n$ of x^* such that $f(x) \geq f(x^*)$ for all $x \in S \cap V$.

Consider $O := S \cap N \cap V$, which is an open neighbourhood of x^* in the subspace topology of $S \cap N$.

Because $P: W \rightarrow S \cap N$ is continuous, $W_0 := P^{-1}(O)$ is an open nbd. of y^* in W (hence open in $\mathbb{R}^{n-m}$).

For any $y \in W_0$, its image $P(y)$ lies in $O \subset S \cap V$.

Therefore, $h(y) = f(P(y)) \geq f(x^*) = f(P(y^*)) = h(y^*)$.

Thus y^* is a local minimum of h on W .

($\Leftarrow$) Assume y^* is an unconstrained local minimum of h on W .

By defⁿ, there exists an open nbhd $W_0 \subset W$ of y^* such that $h(y) \geq h(y^*)$ for all $y \in W_0$.

Because $P: W \rightarrow S \cap N$ is a homeomorphism, it is an open map. This means P maps open sets in W to open sets in the subspace topology of $S \cap N$, so $P(W_0)$ is relatively open in $S \cap N$. Then

$$P(W_0) = S \cap N \cap V$$

for some open set $V \subset \mathbb{R}^n$. Let $N_0 = N \cap V$, so N_0 is an open nbhd of x^* in $\mathbb{R}^n$. For any $x \in S \cap N_0$, we have $x \in P(W_0)$. Because P is bijective, there is a unique $y \in W_0$ such that $x = P(y)$. Then

$$\begin{aligned} f(x) &= f(P(y)) = h(y) \geq h(y^*) \\ &= f(P(y^*)) \\ &= f(x^*). \end{aligned}$$

Thus, x^* is a constrained local minimum of f on S . $\blacksquare$

The Tangent Space

The tangent space to S at x^* is defined as

$$T_{x^*} S := \ker Dg(x^*)$$

Lemma 2: $T_{x^*} S := \ker Dg(x^*)$

$$= \{ (v_y, 0, \phi(y^*) v_y) : v = (v_y, v_z) \in \mathbb{R}^{2-m} \times \mathbb{R}^m \}$$

Proof: By differentiating the identity $g(y, \phi(y)) = 0 \quad \forall y \in W$

we get

$$D_y g(y, \phi(y)) + D_z g(y, \phi(y)) D\phi(y) = 0.$$

Evaluating at y^* , we obtain the derivative of the parametrization

$$D\phi(y^*) = -[D_z g(x^*)]^{-1} D_y g(x^*). \quad \text{--- (1)}$$

A vector $v = (v_y, v_z) \in \mathbb{R}^{n-m} \times \mathbb{R}^m$ lies in $T_{x^*}S$ if and only if $Dg(x^*)v = 0$, if and only if

$$D_y g(x^*) v_y + D_z g(x^*) v_z = 0;$$

if and only if

$$v_z = -[D_z g(x^*)]^{-1} D_y g(x^*) v_y$$

if and only if (using equation (1))

$$v_z = D\phi(y^*) v_y.$$

We have therefore shown that

$$T_{x^*}S := \{ (v_y, D\phi(y^*) v_y) : v_y \in \mathbb{R}^{n-m} \}.$$

Theorem 8.1: (First-order necessary conditions for constrained minimum)
If $x^* = (y^*, z^*)$ is a local minimum of f on S , then there exists a Lagrange multiplier vector $\Delta \in \mathbb{R}^m$ such that $DL(x^*) = 0$, where the Lagrangian

$$L: U \rightarrow \mathbb{R} \text{ is defined as}$$

$$L(x) := f(x) - \Delta^T g(x)$$

Proof: By Lemma 1, y^* is a local minimum for h w.r.p

Since $h \in C^2(W, \mathbb{R})$, $Dh(y^*) = 0$.

By chain rule:

$$Dh(y^*) = D_y f(x^*) + D_z f(x^*) D\phi(y^*) = 0 \quad (2)$$

Define the column vector $\Delta \in \mathbb{R}^m$ by

$$\Delta^T = D_z f(x^*) [D_z g(x^*)]^{-1}.$$

We immediately get

$$D_z f(x^*) - \Delta^T D_z g(x^*) = 0 \quad (3)$$

Substituting (1) into (2) gives us

$$D_y f(x^*) - D_z f(x^*) [D_z g(x^*)]^{-1} D_y g(x^*) = 0.$$

Using our defⁿ for Δ^T gives us
 $D_y f(x^*) - \Delta^T D_y g(x^*) = 0 \quad (4)$

We can write $DL(x^*)$ in block form as

$$DL(x^*) = \begin{bmatrix} D_y L(x^*) & D_z L(x^*) \end{bmatrix}$$

Equations (3) and (4) ~~say~~ state that
 $D_y L(x^*) = 0$ and $D_z L(x^*) = 0$. $\square$

Thus, $DL(x^*) = 0$.

Note that by recalling that
 $\nabla f(x^*) = Df(x^*)^T$
 the equation $DL(x^*) = 0$ can alternatively be
 written as

$$\begin{aligned} \nabla f(x^*) &= Dg(x^*)^T \Delta \\ &= \sum \nabla g_i(x^*) \lambda_i \\ &= \sum \lambda_i \cdot \nabla g_i(x^*) \end{aligned}$$

where $g := \begin{bmatrix} g_1 \\ \vdots \\ g_m \end{bmatrix}$ and $\Delta = \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_m \end{bmatrix}$

Thus, the gradient of f lies in the span
 of the gradients of g at the point x^*

Proposition 1: ^{Assume} f and g are as before, x^* is a regular
 point of S . Then if $DL(x^*) = 0$ where
 $L(x) = f(x) - \Delta^T g(x) \quad (x \in U)$
 the $\Delta \in \mathbb{R}^m$ is uniquely determined by f and g .

Proof: We have $Dg(x^*)^T \Delta = \nabla f(x^*)$.
Put $A := Dg(x^*)$ so the condition is
 $A^T \Delta = \nabla f(x^*)$.

Since x^* is a regular point, A is an $m \times n$ matrix with full row rank m .

This implies that the $m \times m$ matrix

$$B := AA^T$$

is symmetric positive-definite, and therefore invertible.

Multiplying both sides of the condition by A we get

$$B \Delta = A A^T \Delta = A \nabla f(x^*)$$

Therefore,

$$\begin{aligned} \Delta &= B^{-1} A \nabla f(x^*) \\ &= \left[Dg(x^*) \{Dg(x^*)\}^T \right]^{-1} Dg(x^*) \nabla f(x^*). \end{aligned}$$

Thus Δ is uniquely determined. $\square$

What we have shown is that if x^* is a local minimum of f on S , or even if only $DL(x^*) = 0$, then the Lagrangian multiplier vector, and therefore the Lagrangian is unique.

Constrained Local Extrema: 2nd-order conditions

To analyze the 2nd-order behavior of f on S , we must evaluate the 2nd-order derivative $D^2h(y^*)$ of the function h .

We need to make use of the fact that $D^2L(x^*) = 0$, so we need to write h in terms of L .

Indeed, since $g \circ P \equiv 0$, or in other words

$$g(P(y)) = g(y, \phi(y)) = 0 \text{ for all } y \in W,$$

we can write

$$\begin{aligned} h(y) &= f(P(y)) = f(P(y)) - \Delta^T g(P(y)) \\ &= L(P(y)) \text{ for all } y \in W. \end{aligned}$$

Let $w \in \mathbb{R}^{n-m}$ be an arbitrary vector. We examine the real function

$$\begin{aligned} \psi(t) &:= h(y^* + tw) \\ &= L(P(y^* + tw)) \end{aligned}$$

in a neighbourhood of $t=0$.

We first note that the second direction derivative of h at y^* ~~is given by~~ along w is given by

$$D_w^2 h(y^*) = w^T D^2 h(y^*) w.$$

But

$$\begin{aligned} \psi''(0) &= \lim_{\delta \rightarrow 0} \frac{\psi'(\delta) - \psi'(0)}{\delta} \\ &= D_w^2 h(y^*). \end{aligned}$$

Thus,

$$\psi''(0) = \omega^T D^2 h(y^*) \omega.$$

Now we find another expression for $\psi''(0)$.
For t in a ^{small} neighbourhood of 0, define

$$v(t) = \frac{d}{dt} p(y^* + t\omega) \in \mathbb{R}^n.$$

By the chain rule,

$$v(t) = D p(y^* + t\omega) \omega.$$

At $t=0$, we have:

$$\begin{aligned} v^* = v(0) &= D p(y^*) \omega \\ &= (\omega, D\phi(y^*) \omega) \in T_{x^*} S \\ &\quad \text{where } \mathcal{O} = \ker Dg(x^*). \end{aligned}$$

Using the chain-rule, we can compute

$$\begin{aligned} \psi''(t) &= v(t)^T D^2 L(p(y^* + t\omega)) v(t) \\ &\quad + D L(p(y^* + t\omega)) v'(t). \end{aligned}$$

Evaluating at $t=0$ and using the condition that $D L(x^*) = 0$, the term $D L(x^*) v'(0)$ is annihilated. We are left with

$$\begin{aligned} \psi''(0) &= v(0)^T D^2 L(x^*) v(0) \\ &= (v^*)^T D^2 L(x^*) v^*. \end{aligned}$$

We have thus proven the Fundamental Identity of constrained optimization:

If $D L(x^*) = 0$ then

$$\omega^T D^2 h(y^*) \omega = v^T D^2 L(x^*) v$$

for all $\omega \in \mathbb{R}^{n-m}$, where $v = (\omega, D\phi(y^*) \omega) \in T_{x^*} S = \ker Dg(x^*)$.

Theorem 2 (2^{nd} -order Necessary Condition).

If x^* is a constrained local minimum of f on S , then $v^T D^2 L(x^*) v \geq 0$ for all $v \in \ker Dg(x^*)$.

Proof:- If x^* is a constrained local minimum of f on S , then y^* is an unconstrained local minimum of h . Therefore $D^2 h(y^*)$ is positive semi-definite, meaning $w^T D^2 h(y^*) w \geq 0 \quad \forall w \in \mathbb{R}^{n-m}$.

If $v \in \ker Dg(x^*)$ then Lemma 2 implies

$$v = (w, D\phi(y^*)w)$$

for some $w \in \mathbb{R}^{n-m}$. By the Fundamental Identity proved above (~~since~~ which holds since $DL(x^*) = 0$ by Thm 1)

we get $v^T D^2 L(x^*) v = w^T D^2 h(y^*) w \geq 0$. $\square$

Theorem 3 (2^{nd} -order Sufficient Condition).

Suppose ~~the~~ there exists $\Lambda \in \mathbb{R}^m$ such that $DL(x^*) = 0$ [where $L(x) = f(x) - \Lambda^T g(x)$ for $x \in U$].

If $v^T D^2 L(x^*) v > 0$ for all non-zero $v \in \ker Dg(x^*)$ then x^* is a strict constrained local minimum of f on S .

Proof:- By the Fundamental Identity, $w^T D^2 h(y^*) w > 0$ for all non-zero $w \in \mathbb{R}^{n-m}$.

Therefore the unconstrained Hessian $D^2 h(y^*)$ is positive definite. By the corresponding theorem for the unconstrained case and the fact that $h = L \circ P$ we get

$$\begin{aligned} Dh(y^*) &= DL(P(y^*)) DP(y^*) \\ &= DL(x^*) DP(y^*) = 0. \end{aligned}$$

Therefore, y^* is a strict unconstrained local ~~minima~~ minimum of h . Thus x^* is a strict constrained local minimum of f on S . $\square$

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Theorem 1 (Implicit Function Theorem, Local Homeomorphism Version). *Let $U \subset \mathbb{R}^n$ be an open set, and let $g : U \rightarrow \mathbb{R}^m$ be a C^2 function, where $1 \leq m < n$. Write $x \in \mathbb{R}^n$ as $x = (y, z)$, where $y \in \mathbb{R}^{n-m}$ and $z \in \mathbb{R}^m$. Suppose $x^* = (y^*, z^*) \in U$ satisfies $g(y^*, z^*) = 0$, and the $m \times m$ partial Jacobian matrix $D_z g(y^*, z^*)$ is invertible. Define the constraint surface $S := \{x \in U \mid g(x) = 0\}$.*

Then there exists an open neighbourhood $W \subset \mathbb{R}^{n-m}$ of y^ , an open neighbourhood $N \subset \mathbb{R}^n$ of x^* , and a unique C^2 function $\phi : W \rightarrow \mathbb{R}^m$ such that the parametrization mapping $P : W \rightarrow S \cap N$ defined by $P(y) := (y, \phi(y))$ is a homeomorphism.*

Setup

Fix $f \in C^2(U, \mathbb{R})$ where U is open in $\mathbb{R}^n$. Let $g : U \rightarrow \mathbb{R}^m$ be of class C^2 , with $m < n$. Define $S := \{x \in U \mid g(x) = 0\}$. We seek the local extrema of f restricted to S .

Establishing suitable necessary and sufficient conditions requires a local parametrization of the constraint manifold S . We therefore assume henceforth that $x^* \in S$ is a regular point of g . This means that the $m \times n$ Jacobian matrix $Dg(x^*)$ has full rank m .

By permuting the coordinates of $x = (y, z) \in \mathbb{R}^{n-m} \times \mathbb{R}^m$, we may assume without loss of generality that the $m \times m$ partial Jacobian $D_z g(x^*)$ is invertible.

We can now apply the Inverse Function Theorem exactly as stated above to obtain $y^* \in W \subset \mathbb{R}^{n-m}$, $x^* \in N \subset \mathbb{R}^n$, $\phi \in C^2(W, \mathbb{R}^m)$, and a homeomorphism $P : W \rightarrow S \cap N$ defined by $P(y) = (y, \phi(y))$. Observe that the theorem's conditions imply $\phi(y^*) = z^*$ and $P(y^*) = x^*$.

We now show a strict correspondence between the constrained local minimum of f on S and the unconstrained local minimum of h on W , where we define the reduced function $h : W \rightarrow \mathbb{R}$ by $h(y) := f(P(y)) = f(y, \phi(y))$.

Lemma 1. *x^* is a constrained local minimum of f on S if and only if y^* is an unconstrained local minimum of $h = f \circ P$ on W .*

Proof. ($\implies$) Assume x^* is a constrained local minimum of f on S . Then there exists an open neighbourhood $V \subset \mathbb{R}^n$ of x^* such that $f(x) \geq f(x^*)$ for all $x \in S \cap V$.

Consider $O := S \cap N \cap V$, which is an open neighbourhood of x^* in the subspace topology of $S \cap N$. Because $P : W \rightarrow S \cap N$ is continuous, $W_0 := P^{-1}(O)$ is an open neighbourhood of y^* in W (and hence open in $\mathbb{R}^{n-m}$).

For any $y \in W_0$, its image $P(y)$ lies in $O \subset S \cap V$. Therefore,

$$h(y) = f(P(y)) \geq f(x^*) = f(P(y^*)) = h(y^*).$$

Thus, y^* is a local minimum of h on W .

($\impliedby$) Assume y^* is an unconstrained local minimum of h on W . By definition, there exists an open neighbourhood $W_0 \subset W$ of y^* such that $h(y) \geq h(y^*)$ for all $y \in W_0$.

Because $P : W \rightarrow S \cap N$ is a homeomorphism, it is an open map. This means P maps open sets in W to open sets in the subspace topology of $S \cap N$. Thus, $P(W_0)$ is relatively open in $S \cap N$, meaning $P(W_0) = S \cap N \cap V$ for some open set $V \subset \mathbb{R}^n$.

Let $N_0 = N \cap V$; since N and V are open in $\mathbb{R}^n$, N_0 is an open neighbourhood of x^* in $\mathbb{R}^n$. For any $x \in S \cap N_0$, we have $x \in P(W_0)$. Because P is bijective, there is a unique $y \in W_0$ such that $x = P(y)$. Then,

$$f(x) = f(P(y)) = h(y) \geq h(y^*) = f(P(y^*)) = f(x^*).$$

Thus, x^* is a constrained local minimum of f on S . □

The Tangent Space

The tangent space to S at x^* is defined as $T_{x^*}S := \ker Dg(x^*)$.

Lemma 2. $T_{x^*}S = \{(v_y, D\phi(y^*)v_y) \mid v_y \in \mathbb{R}^{n-m}\}$.

Proof. By differentiating the identity $g(y, \phi(y)) = 0$ with respect to y for all $y \in W$, we apply the chain rule to get:

$$D_y g(y, \phi(y)) + D_z g(y, \phi(y))D\phi(y) = 0.$$

Evaluating at y^* , we obtain the derivative of the parametrization:

$$D\phi(y^*) = -[D_z g(x^*)]^{-1}D_y g(x^*). \quad (1)$$

A vector $v = (v_y, v_z) \in \mathbb{R}^{n-m} \times \mathbb{R}^m$ lies in $T_{x^*}S$ if and only if $Dg(x^*)v = 0$, which holds if and only if

$$D_y g(x^*)v_y + D_z g(x^*)v_z = 0.$$

Isolating v_z , this is equivalent to:

$$v_z = -[D_z g(x^*)]^{-1}D_y g(x^*)v_y.$$

Using equation (1), this perfectly simplifies to:

$$v_z = D\phi(y^*)v_y.$$

We have therefore shown that $T_{x^*}S = \{(v_y, D\phi(y^*)v_y) \mid v_y \in \mathbb{R}^{n-m}\}$. □

First-Order Necessary Conditions

Theorem 2. *If $x^* = (y^*, z^*)$ is a constrained local minimum of f on S , then there exists a Lagrange multiplier vector $\lambda \in \mathbb{R}^m$ such that $DL(x^*) = 0$, where the Lagrangian $L : U \rightarrow \mathbb{R}$ is defined as $L(x) := f(x) - \lambda^\top g(x)$.*

Proof. By Lemma 1, y^* is a local minimum for $h = f \circ P$. Since $h \in C^2(W, \mathbb{R})$, its unconstrained derivative must vanish: $Dh(y^*) = 0$. By the chain rule:

$$Dh(y^*) = D_y f(x^*) + D_z f(x^*)D\phi(y^*) = 0. \quad (2)$$

Define the column vector $\lambda \in \mathbb{R}^m$ by

$$\lambda^\top := D_z f(x^*) [D_z g(x^*)]^{-1}.$$

Multiplying on the right by $D_z g(x^*)$ immediately yields:

$$D_z f(x^*) - \lambda^\top D_z g(x^*) = 0. \quad (3)$$

Substituting (1) into (2) gives us:

$$D_y f(x^*) - D_z f(x^*) [D_z g(x^*)]^{-1} D_y g(x^*) = 0.$$

Using our definition for $\lambda^\top$, this simplifies to:

$$D_y f(x^*) - \lambda^\top D_y g(x^*) = 0. \quad (4)$$

We can write $DL(x^*)$ in block form as $DL(x^*) = [D_y L(x^*) \quad D_z L(x^*)]$. Equations (4) and (3) respectively state that $D_y L(x^*) = 0$ and $D_z L(x^*) = 0$. Thus, $DL(x^*) = 0$.

Note that by recalling $\nabla f(x^*) = Df(x^*)^\top$, the equation $DL(x^*) = 0$ can alternately be written as:

$$\nabla f(x^*) = Dg(x^*)^\top \lambda = \sum_{i=1}^m \lambda_i \nabla g_i(x^*),$$

where $g := \begin{bmatrix} g_1 \\ \vdots \\ g_m \end{bmatrix}$ and $\lambda = \begin{bmatrix} \lambda_1 \\ \vdots \\ \lambda_m \end{bmatrix}$. Thus, the gradient of f lies strictly in the span of the gradients of g . □

Proposition 1. *Assume f and g are as before, and x^* is a regular point of S . If $DL(x^*) = 0$ where $L(x) = f(x) - \lambda^\top g(x)$ for $x \in U$, then the multiplier $\lambda \in \mathbb{R}^m$ is uniquely determined by f and g .*

Proof. We have $Dg(x^*)^\top \lambda = \nabla f(x^*)$. Let $A := Dg(x^*)$, so the condition becomes $A^\top \lambda = \nabla f(x^*)$.

Since x^* is a regular point, A is an $m \times n$ matrix with full row rank m . This implies that the $m \times m$ matrix $B := AA^\top$ is symmetric positive-definite, and therefore invertible.

Multiplying both sides of the condition on the left by A , we get:

$$AA^\top \lambda = A \nabla f(x^*).$$

Isolating λ yields:

$$\begin{aligned} \lambda &= B^{-1} A \nabla f(x^*) \\ &= [Dg(x^*) \{Dg(x^*)\}^\top]^{-1} Dg(x^*) \nabla f(x^*). \end{aligned}$$

Thus, λ is uniquely determined. □

We have shown that if x^* is a local minimum of f on S , or even if we simply have a stationary point where $DL(x^*) = 0$, the regularity condition ensures the Lagrange multiplier vector is strictly unique.

Constrained Local Extrema: 2nd-order conditions

To analyze the second-order behavior of f on S , we must evaluate the second-order derivative $D^2h(y^*)$ of the unconstrained function h . We need to make use of the fact that $DL(x^*) = 0$, so we write h purely in terms of L .

Indeed, since $g \circ P \equiv 0$ on W (meaning $g(P(y)) = 0$ for all $y \in W$), we can write:

$$h(y) = f(P(y)) = f(P(y)) - \lambda^\top g(P(y)) = L(P(y))$$

for all $y \in W$.

Let $w \in \mathbb{R}^{n-m}$ be an arbitrary vector. We examine the real-valued function:

$$\psi(t) := h(y^* + tw) = L(P(y^* + tw))$$

in a neighbourhood of $t = 0$.

We first note that the second directional derivative of h at y^* along w is given by:

$$D_w^2 h(y^*) = w^\top D^2 h(y^*) w.$$

But by definition,

$$\psi''(0) = \lim_{\delta \rightarrow 0} \frac{\psi'(\delta) - \psi'(0)}{\delta} = D_w^2 h(y^*).$$

Thus,

$$\psi''(0) = w^\top D^2 h(y^*) w.$$

Now we find another expression for $\psi''(0)$. For small t in a neighbourhood of 0, define:

$$v(t) = \frac{d}{dt} P(y^* + tw) \in \mathbb{R}^n.$$

By the chain rule,

$$v(t) = DP(y^* + tw)w.$$

At $t = 0$, we have:

$$v^* = v(0) = DP(y^*)w = (w, D\phi(y^*)w) \in \ker Dg(x^*).$$

Using the chain rule, we can compute the second derivative of the composition:

$$\psi''(t) = v(t)^\top D^2 L(P(y^* + tw))v(t) + DL(P(y^* + tw))v'(t).$$

Evaluating at $t = 0$ and using the condition that $DL(x^*) = 0$, the term containing $v'(0)$ is completely annihilated. We are left with:

$$\psi''(0) = v(0)^\top D^2 L(x^*)v(0) = (v^*)^\top D^2 L(x^*)v^*.$$

We have thus proven the **Fundamental Identity of Constrained Optimization**:
If $DL(x^*) = 0$, then

$$w^\top D^2 h(y^*)w = v^\top D^2 L(x^*)v$$

for all $w \in \mathbb{R}^{n-m}$, where $v = (w, D\phi(y^*)w) \in T_{x^*}S = \ker Dg(x^*)$.

Theorem 3 (Second-Order Necessary Condition). *If x^* is a constrained local minimum of f on S , then $v^\top D^2 L(x^*)v \geq 0$ for all $v \in \ker Dg(x^*)$.*

Proof. If x^* is a constrained local minimum of f on S , then y^* is an unconstrained local minimum of h . Therefore, the unconstrained Hessian $D^2h(y^*)$ is positive semi-definite, meaning $w^\top D^2h(y^*)w \geq 0$ for all $w \in \mathbb{R}^{n-m}$.

If $v \in \ker Dg(x^*)$, then Lemma 2 implies $v = (w, D\phi(y^*)w)$ for some $w \in \mathbb{R}^{n-m}$. By the Fundamental Identity proved above (which holds since $DL(x^*) = 0$ by Theorem 1), we get:

$$v^\top D^2L(x^*)v = w^\top D^2h(y^*)w \geq 0.$$

□

Theorem 4 (Second-Order Constrained Sufficient Condition). *Suppose there exists $\lambda \in \mathbb{R}^m$ such that $DL(x^*) = 0$, where $L(x) = f(x) - \lambda^\top g(x)$ for $x \in U$. If $v^\top D^2L(x^*)v > 0$ for all non-zero $v \in \ker Dg(x^*)$, then x^* is a strict constrained local minimum of f on S .*

Proof. We evaluate the first and second derivatives of the unconstrained function h . First, because $h = L \circ P$, the chain rule gives:

$$Dh(y^*) = DL(P(y^*))DP(y^*) = DL(x^*)DP(y^*).$$

Since we are given $DL(x^*) = 0$, it immediately follows that $Dh(y^*) = 0$, establishing that y^* is a stationary point of h .

Second, by the Fundamental Identity, our assumption implies that for all non-zero $w \in \mathbb{R}^{n-m}$, the associated tangent vector $v = (w, D\phi(y^*)w)$ is non-zero, and thus:

$$w^\top D^2h(y^*)w = v^\top D^2L(x^*)v > 0.$$

Therefore, the unconstrained Hessian $D^2h(y^*)$ is strictly positive-definite.

By the corresponding theorem for the unconstrained case, since $Dh(y^*) = 0$ and $D^2h(y^*) \succ 0$, y^* is a strict unconstrained local minimum of h . By Lemma 1, this implies x^* is a strict constrained local minimum of f on S . □